

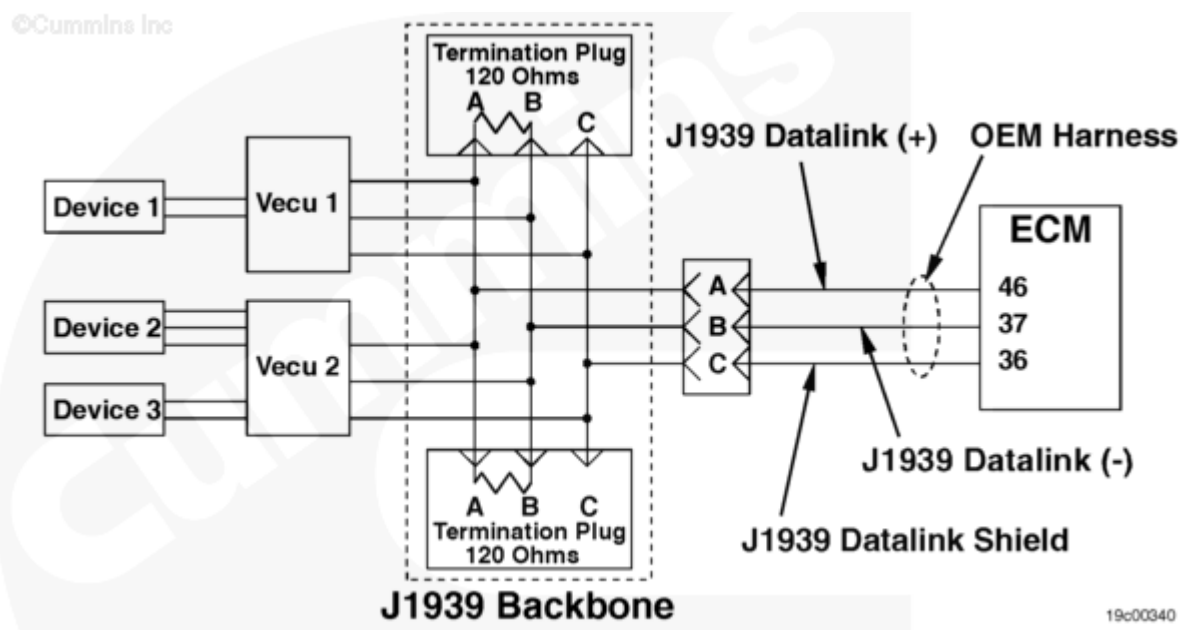
Fault Code: 3714

Engine Protection Torque Derate - Condition Exists

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 3714 PID(P): SPN: FMI: 11/31 Lamp: Amber SRT:	Engine Protection Torque Derate - Condition Exists. SAE J1939 multiplexing heartbeat message from OEM device lost or torque derate imposed by OEM device.	Amber warning lamp will illuminate. Torque derate will be imposed.



J1939 Data Link Multiplexing Circuit

Circuit Description

Normally, switches, accelerators, and other components are connected to the engine electronic control module (ECM) directly through individual wires. Multiplexing allows those same components to be hardwired to an OEM electronic control unit (ECU) in the cab. Then component values and states from components such as sensors, accelerators, and switches can be transmitted from the OEM device to the Cummins® engine ECM over the SAE J1939 data link.

Messages sent from OEM device is received by the Cummins® engine ECM and used for controlling the engine. The Cummins® ECM and OEM device **must** be configured properly so that proper operation of the multiplexed components will occur.

Component Location

The engine ECM is located on the intake side of the engine.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The ECM does **not** receive a valid J1939 heartbeat message from the multiplexed OEM device for more than 60 seconds.

The multiplexed OEM device sends a request for a torque derate.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light 30 seconds after the heartbeat message from the OEM device is lost or a torque derate is requested by the OEM device.
- The ECM will impose a torque derate.

Conditions For Clearing The Fault Code

The ECM will turn off the amber CHECK ENGINE light 30 seconds after the heartbeat message from the OEM device is restored or a torque derate is no longer being requested by the OEM device.

Shoptalk

Verify the ECM calibration is correct. Check the calibration revision history found on QuickServe™ Online for applicable fixes to the calibration stored in the ECM. If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)

When this fault code is active, some multiplexed devices may **not** function as desired.

This fault can occur for the following reasons:

- When the ECM is setup properly (components enabled and OEM device source addressed correctly) to receive information from an OEM device, but the OEM device is actually transmitting a message that says that component is **not** available for multiplexing. This can be caused when a multiplexed component is enabled in the Cummins® ECM, but the OEM device source address of the device transmitting the component message is incorrect in the

Cummins® ECM, or the OEM device is **not** setup to transmit the multiplexed component message.

- A malfunctioning J1939 data link connection between the OEM device and Cummins® ECM, a malfunctioning connection between the component and the OEM device, a malfunctioning OEM device, or a malfunctioning Cummins® ECM. It can be necessary to contact the OEM for the proper multiplexing configuration.
- When the heartbeat message from the OEM device is lost or interrupted. In addition to the fault, a 60 percent torque derate will be imposed. Refer to the OEM service manual.
- When the OEM device is requesting a torque derate between 0 and 100 percent. Refer to the OEM service manual.

Refer to Troubleshooting Fault Code t05-3714 (</qs3/pubsys2/xml/en/procedures/82/82-t05-3714.html>)

Last Modified: 02-Sep-2010
